

Pedodontics and Dental Public Health

Departmental Objectives (Pedodontics):

At the end of the course, the students should be able to:

- Diagnose and manage dental diseases of pediatric patients
- Treat oral diseases for infants and children through adolescence, including those of all ages with special care needs (e.g. handicapped).
- Motivate and treat psychologically challenged pediatric dental patients
- Manage the disabled children effectively and efficiently to the needs of individual requirement and conditions
- Assess and refer pediatric patients to different specialties appropriately
- Resuscitate pediatric patients and manage emergencies
- Promote infants' and children's healthy feeding practice
- Describe the impact of socio-economic and cultural background on initiation and progression of dental diseases in children
- Rehabilitate the patient with a healthy permanent dentition
- Counsel and motivate children and parents about oral hygiene, encourage to have a healthy oral cavity
- Instill a positive attitude and behavior to maintain a good oral health
- Utilize the basic concept of children dentistry with its preventive aspect in dental practice

List of Competencies to acquire (Pedodontics):

- Washing hands
- Wearing of gloves and glasses
- Identify primary and permanent dentition
- Identify dental and surgical instruments, materials
- History taking and clinical examination in the dental chair
- Diagnose and manage of gingival and periodontal diseases, acute viral infections
- Diagnose and manage soft tissue lesions
- Diagnose and refer pre-malignant and malignant lesions
- CLI-VII cavity preparation
- Anterior strip and stainless steel crowning
- Diagnose and manage patients with developmental, hereditary and chromosomal disorders
- Diagnose and manage expression of nutritional deficiencies in dentitions
- Diagnose, manage and refer patients with communicative disorders
- Oro-facial pain and infection control with the help of local anaesthesia, relative anaesthesia, antibiotics and other drugs
- Manage types of pulpal pathologies
- Interpret radiographs used in pediatric dentistry
- Manage traumatized anterior segment of dentition
- Apply fissure sealants and fluorides
- Reshape fissures and perform prophylactic odontology
- Apply local anesthesia
- Demonstrate knowledge of relative analgesia

- Extract primary and permanent teeth of children
- Describe the techniques of frenectomy, operculectomy in children
- Describe surgical techniques of removing ranula, epulis and small cysts
- Perform Hand scaling
- Describe the concept of kiddy partial denture
- Describe the concept of fixed and removable space maintainer
- Teach the pediatric patients about brushing techniques and oral hygiene

Departmental Objectives (Dental Public Health):

At the end of the course, the students should be able to:

- Demonstrate a complete understanding about the concepts of Public health and primary health care and their delivery system in the community .
- Describe different preventive measures to address the prevailing oral and dental problems in the community in each and every level.
- Describe the concept and methods of epidemiology to design simple research in dental background
- Apply concept of behavioural science to organize oral health education program to motivate children , parents, special care group & help them to have a healthy oral cavity
- Describe the concept of dental jurisprudence and address the legal issues in dentistry

List of Competencies to acquire (Dental Public Health):

- Plan & conduct field survey properly
- Provide instruction about—Brushing, Flossing, Inter-dental cleansing aid, Mouthwash
- Produce & promote oral health promotional materials such as poster, festoon, leaflet, hand bill, banner etc.
- Interact with the community people about oral health status.
- Rehabilitate the peoples with a healthy permanent dentition by delivering appropriate knowledge of prevention.
- Act as a social worker regarding establishing a good oral health status among the community people.
- Instruction about proper oral hygiene practice.
- Act as a oral health educator.
- Interpret research findings from published journals and articles.
- Students competent to work in primary health care settings
- Practice professional life ethically and according to the legal framework of the country.
- Demonstrates the knowledge understanding of social and psychological pattern of
- Diseases causation and its management accordingly.

Distribution of teaching-learning hours of Pedodontics and Dental Public Health:

Lecture	Tutorial	Practical+ Clinic Demonstrative works/Field visit	Total Teaching hours (including clinical rotation)	Integrated teaching (Common)	Study tour	Formative Exam		Summative exam	
						Preparatory leave	Exam time	Preparatory leave	Exam time
84 hrs	37	150hrs	271hrs	10 hrs	10 days	10 days	20 days	10 days	35 days
53 hrs	107 hrs	7hrs 5hrs (Field visit)	172 hrs						
137hrs	144hrs	162 hrs	443hrs						

Group-A

Teaching - learning methods, teaching aids and evaluation (Pedodontics)

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture (video, Presentation)	Chair-side teaching (OPD), Tutorials, O.T. demonstration. O.H. clinic	Assignment, group & self study	Multimedia, OHP, White board, laptop, desktop, slide projector, video, dummy, pathology report, X-ray plate, view box, model, TV, specimen etc	Item exam, Card final, Term exam, class performance, Term final (written, oral, practical, clinical)

Group-B

Teaching/learning methods, teaching aids and evaluation (Dental Public Health)

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture, Integrated teaching	Tutorial Practical/Field visit	Assignment, self assessment & self study.	Computer & Multimedia Chalk & board White board & markers OHP Slide projector Flip Chart Models Specimens projector Study guide & Hand out	Item examination (oral) Practical item examination(Oral & practical) Card completion Examination (Written) Term final Examination(Written, oral & practical)

Professional Examination:

Marks Distribution of assessment of Professional Examination

Total Marks: 300

- Written 100 : Group A : Pedodontics 50 + Group B: Dental Public Health 50) (SAQ=35+35) + (MCQ=10+10)+ (Formative assessment mark = 5+5)
- SOE 100 : Board A: Pedodontics 50 + Board B: Dental Public Health 50)
- OSPE/Practical : Pedodontics 50 + Dental Public Health 50

Learning Objectives and Course Contents in Pedodontics

Learning Objectives	Contents	Teaching Hours
- Students will be able to - Describe the concept of paediatric dentistry - State the importance to save the primary and permanent dentition - Take history of paediatric dental patient - Differentiate between infants, toddler, child, teenage and adolescent	1. Introduction to children dentistry & different branches of paediatric dentistry - Definition - Scope - Aims and Objectives - Importance - History & Notation	L-3hrs T-1hrs
- Describe facial development - Identify primary & permanent teeth, - State the chronology of teeth (shedding, eruption and calcification) - Differences between deciduous and permanent teeth. - Know the Development of dentition from birth to adolescence.	2. Dento-facial development, chronology & morphology of primary and permanent dentition - Definitions of growth & development - Importance - Dento-facial development (Maxilla and Mandible) and age related changes - Chronology & morphology of primary & permanent teeth	L-5, P+C+D-2, T-2hrs
- State the principles of history taking, examination and investigation - Perform clinical examination, diagnosis and treatment planning	3. History, diagnosis & treatment planning for paediatric patients in the dental chair - Case history recording - Diagnosis - Treatment planning	L-4, P+C+D-5, T-1hrs
- State etiology, pathology, classify & manage different periodontal & gingival diseases in primary and permanent dentition - Perform hand scaling - Prevent gingival & periodontal diseases	4. Gingival & periodontal diseases in children - Features and periodontal condition Features of normal gingiva and periodontium in children - Definition, classification, Etiology, Pathogenesis and management of gingival seen in children and adolescents - Normal periodontium in children - Gingival diseases in children - Periodontal diseases in children	L-5, P+C+D-10, T-2hrs

Learning Objectives	Contents	Teaching Hours
Students will able to ● Define, classify and manage different soft tissue lesions diseases including rashes, ulcers and acute and chronic viral manifestation in the oral cavity and face	5. Diseases of oral mucous membrane, soft tissues & other supporting structure ● Aetiological factors and differential diagnosis ● Investigations ● Relationship to systemic disease ● Relevant pharmacology and therapeutics ● Clinical features of the disease	L-5, P+C+D-5, T-1hrs
● Diagnose, classify, know the choice of treatments and refer to the respective departments	6. Pre-cancerous & cancerous lesions commonly found in children ● Aetiological factors and differential diagnosis ● Investigations ● Relationship to systemic disease ● Relevant pharmacology and therapeutics ● Clinical features of the disease	L-3, P+C+D- 2hrs
● Describe the factors causing the disorders ● Define and manage different developmental, hereditary and chromosomal anomalies found in dental tissues ● Mention the clinical features , management and the treatment plan of the patients with anomalies in dental tissues	7. Congenital abnormalities in children ● Definition, ● Classification ● Developmental, hereditary & chromosomal disorders in children	L-3, P+C+D- 5, T-2hrs
● Classify and manage different pathological conditions found in children due to vitamin nutritional deficiencies & hormonal imbalance ● Diagnose & treat the diseases	8. Nutritional and hormonal factors in dentistry ● Age and sex and race, Cause, Type ● Factors (local and systemic or genetic) ● Endocrine Disorder	L-3, T-2hrs

Learning Objectives	Contents	Teaching Hours
Students will able to ● Apply Psychology principles in management of child patients in the dental office ● Describe psychological development from birth through teenage ● Manage Dental fear, anxiety ● Provide non- Pharmacologic and Pharmacologic management of behavior	9. Behavioral management, child psychology ● Definition ● Classification and types of behavior ● Factors influencing child behavior ● Importance of understanding Child Psychology in Pedodontics ● Child abuse and neglect	L-4, P+C+D-5, T- 1hrs
● Mention classifications and compositions of local anaesthesia, different techniques and application of L.A., side effects and complications of L.A. ● Mention R.A. & G.A. ● Mention classification, pharmacology, doses and administrations of antibiotics and other common drugs ● Describe analgesics, anti-inflammatory and antibiotics commonly prescribed for children ● Describe conscious sedation including nitrous oxide- oxygen inhalational anaesthesia	10.Pain and infection management in paediatric dentistry ● Local anaesthesia ● Relative analgesia ● General anaesthesia ● Antibiotics and other essential drugs ● Pharmacological principles in Pediatric Dentistry- drug dosage formulae	L-6, P+C+D- 10, T- 4hrs
● State the Causes, perform examinations, diagnosis and provide preventive measures to avoid trauma to the A.S., ● Provide Immediate management , intermediate & final treatment of traumatized, avulsed and fractured teeth, in deciduous dentition and young permanent teeth ● State Sequelae and reaction of teeth to trauma following primary teeth ● Bleaching non vital teeth	11.Traumatic injuries of teeth & their Management(The care of injuries to the anterior segment of teeth) ● Definition, classification, etiology ● Treatment & management ● Prevention of trauma: mouth protectors	L-5, P+C+D- 13, T- 4hrs

Learning Objectives	Contents	Teaching Hours
● State Principles & Diagnosis ● Manage pulpal involved primary, young permanent and permanent teeth including materials used and techniques followed: ○ Pulp capping– direct & indirect. ○ Pulpotomy ○ Pulpectomy ○ Apexogenesis ○ Apexification ● Describe obturation Techniques & material used for primary, young permanent & Permanent teeth ● Describe operative procedures ● Perform manipulations and restorations of decayed primary, young permanent and permanent teeth in children using various restorative materials like Glass Ionomer, Composites & Silver Amalgam. ● Preform Stainless steel, Polycarbonate & Resin Crowns ● Diagnose and prepare CL. I-VII cavities & aesthetic restorations ● Various Isolation Techniques	12. Pediatric Endodontics and Operative Dentistry ● Principles of Pediatric Endodontics and Operative Dentistry ● Classification of Pulpal Pathology in primary, young permanent & permanent teeth. ● Properties of Restorative materials & techniques ● Modifications required for cavity preparation in primary and young permanent teeth. ● Techniques of Isolation ● Atraumatic / Alternative Restorative Technique (ART) ● Crowns: Stainless steel, polycarbonate and anterior strip crowns	L-8, P+C+ D-20, T-4hrs
● Perform drainage of infective lesions ● Extract primary, permanent, supernumerary and buried teeth, surgical exposor of maxillary cuspid ● Perform all the minor oral surgeries ● Manage infections	13. Oral surgery in children ● Indications & contraindications of extractions of primary and permanent teeth in children ● Extraction techniques in children ● Minor oral surgical procedures in children ● Infections control ● Operculectomy, frenectomy, Surgical removal of ranula, epulis and small cysts	L-4, P+C+ D- 20, T-2hrs

Learning Objectives	Contents	Teaching Hours
Describe the procedure of making removable kiddy partial denture	14. Paediatric prosthodontics	L-2, P+C+D-5, T-2hrs
- Describe the problems encountered during primary and mixed dentition phases & provide their management - Construct removable & fixed space maintainers, correct of oral bad habits, narrow upper jaw & crossbites, demonstrate techniques of serial extractions - Mention Clinical features of deleterious oral habits including non- nutritive sucking, mouth breathing, non-functional grinding, masochistic and occupational habits - Manage of oral habits in children	15. Preventive & Interceptive Orthodontics - Oral Habits in children - Definitions, classification and etiology of all habits - Preventive measures - Developing occlusion & its management - Space regainers - Serial extractions. - Space management. - Space maintainers: definition, classification, indications and contra indications, advantages and disadvantages including construction of fixed space maintainers	L-2, P+C+D-5, T-2hrs
- Describe intraoral radiographic techniques - Describe extra oral/specialized radiographic techniques - Describe modification in radiographic techniques in children - Describe radiographic survey in children - Interpret of radiographs used for children, - Mention radiation hazards & protections	I6. Radiology in children - Definition, - Types of radiographs - Various techniques of radiograph	L-2, P+C+D-5, T-2 hrs
- Mention factors responsible (etiology) for caries initiation, and state classifications, pathology and epidemiology with socio-economic factors - Manage early childhood caries including nursing bottle caries - Manage rampant caries - Explain role of food in dental caries and aware parents to apply the knowledge - Apply prophylactic & operative techniques in dental caries prevention by fissure sealants, prophylactic odontomy, reshaping of fissures, application of fluorides - Provide primary care & oral hygiene education in childhood	17. Dental caries: - Definition, Theories and Etio pathogenesis - Epidemiology - Early childhood caries including Nursing bottle caries - definition, classification, etiology, pathogenesis, clinical features, complications and management - Rampant caries -definition, classification, etiology, pathogenesis, clinical features, complications and management - Diet & Dental caries including diet counseling, diet and modifications - Caries activity, caries prediction and caries susceptibility & heir clinical application	L-10, P+C+D-25, T-2hrs

Learning Objectives	Contents	Teaching Hours
● Recognize & manage medical emergencies initially that may occur during dental procedures	18. Medical and Dental emergencies in children and their management	L-5 P+C+D-5hrs
● Manage the patients with mental retardation ● Manage the patients with Down's syndrome, cerebral palsy, epilepsy etc. ● Manage the patient with learning disability such as deafness, blindness etc. ● Treat the dental diseases of medically compromised children ● Manage behaviourally challenged and handicapped children with dental diseases	19. Dental care of children with special needs ● Definition, etiology, clinical features of ○ Physically and Mentally Handicapped Children ○ Medically compromised children ○ Different genetic disorders found in children ● Management of handicapped children in the dental chair with special consideration of communicative disorders(A.S.D) ● Parents' counseling	L-5, P+C+D-8, T-3hrs

Learning Objectives and Course Contents in Dental Public Health

Learning Objectives	Contents	Teaching Hours
Student will be able ● Define health ● Describe history of public health ● Describe history of dental public health ● Explain iceberg of disease ● Describe history of disease ● Explain epidemiological triad ● Describe phase of disease and illness and mention factors affecting these states ● Mention indicator of health ● Describe determinates of health ● Explain dimensions of health	Fundamentals of dental public health Concept of ● Health & Disease ● Determinates of health ● Different dimensions of health ● Public Health and in relating to dentistry	L= 2 T= 4
● Define epidemiology ● Mention the aim and objectives of epidemiology ● Define quarantine, Epidemic, Endemic & Pandemic, Sporadic ● Describe epidemiological study design ● Define screening ● Define & Explain ○ Incidence ○ Prevalence ○ Rate ○ Ratio ○ Proportion ● Epidemiology of dental diseases	● General & Dental Epidemiology	L=3 T=6

Learning Objectives	Contents	Teaching Hours
● Define and Classify of research ● Mention characteristics of research question ● Explain ethics in Research ● Steps of research ● Bias ● Blinding	● Research Methodology	L=2 T=3
● Define Biostatistics ● Explain Concept of Data ● Define & Classify variables ● Describe data collection, compilation, organization analysis, interpretation & presentation of data ● State measures of central tendency ● State measure of dispersion ● Describe normal Distribution curve	● Biostatics ● Data management	L=2 T=6 P=1
● Define sample and population ● Mention types of sampling ● Explain sampling technique	● Sampling methods	L=2 T=4
● Define & mention types of Index ● State importance of Index ● Mention properties of Index ● Use of Index ● Scoring of DMFT (demonstration)	● Index	L=2 T=3
● Define dental ancillary ● Classify ancillary ● State function of ancillary ● Explain rationale of dental ancillary	● Dental Manpower and ancillary	L=1 T=3

Learning Objectives and Course Contents in Dental Public Health & Community Dentistry

Learning Objectives	Contents	Teaching Hours
- State Alma Ata declaration - Define Primary Health Care (PHC) - Explain principles of PHC - Mention core activities of PHC	Primary Health Care	L=2 T=4
- Define planning and its stages in Health Care, Purposes of planning - Evaluation in health care - Mention health care organizations and resources	Planning and evaluation of health care in dentistry	L=2 T=5
- Explain importance & objectives of School dental Health education - A School dental health education program - Planning a school health education program - Describe preventive dietary program - Describe pit and fissure sealant program - Conduct classroom tooth brushing program - Demonstration of fluoride application	School dental health program	L=2 T=4
- Define health promotion & mention its fundamental principles - Define health education and its objectives & principles - Describe different media/levels used in dental health education	- Oral health promotion - Dental Health education	L=1 T=3 P=1
Application of computers in dental health care .	Computer application in dentistry	L=1 T=1

Learning Objectives and Course Contents in Preventive Dentistry

Learning Objectives	Contents	Teaching Hours
Student will be able - Define Prevention - Describe levels of prevention - Describe measures of prevention in relation to dental disease	Concept of Prevention in dentistry	L=1 T=2
- Define dental caries - Classify Dental caries - Epidemiology of caries - List etiological factors of dental caries - Describe theory of dental caries development - Describe prevention & measures in dental caries - Mention uses of Fluoride - Define diet & dietary counseling - ART (Definition, Principle, advantage, disadvantage & technique)	Prevention & Control of Dental Caries	L=3 T=6 P=3
- Define Periodontal disease - Classify periodontal disease - Epidemiology of Periodontal disease - Describe preventive measures - Describe control of periodontal diseases by mechanical, chemotherapeutic & others. - Nutritional effect on periodontal disease	Prevention & Control of Periodontal Disease	L=2 T=4 P=2

Learning Objectives	Contents	Teaching Hours
- Define dental trauma - Classify dental trauma - Describe prevention measures of dental trauma - Explain primary, Secondary & Tertiary prevention - Explain mouth guard	Prevention of Dental Trauma	L=2 T=3
- Define Handicap & dental handicap - Classify Handicap - Mention causative factors of physical , mental, social handicap - Describe management of handicap patient - Explain maintenance of oral hygiene in case of handicap patients	Dental management of Handicap	L=1 T=2
- Define Dental Management of Geriatric People - Classify Dental Management of Geriatric People - Describe preventive measures of oral problem in geriatric population - List influencing factors for using dental care by geriatric people	Dental Management of Geriatric People	L=1 T=2
- Define oral cancer - Types - Common sites - Name pre malignant lesion - Epidemiology of Oral cancer - State etiological factors & preventive measures - Screening technique in diagnosis of oral cancer	Prevention of Oral Cancer & management of Risk factors	L=2 T=3

Learning Objectives and Course Contents in Preventive Dentistry

Learning Objectives	Contents	Teaching Hours
● Define malocclusion ● Classify Malocclusion ● Mention causes of Malocclusion ● Describe prevention & interceptive orthodontics measures ● Identify abnormal oral habits & managements	● Prevention of Malocclusion	L=2 T=4
● Define sterilization ● Classify sterilization ● Describe Preventive of occupational hazards in dentistry ● Define occupational hazard ● List occupational hazards in dentistry ● Control of HIV AIDS, Hepatitis B C & others Bacterial and fungal infection ● Waste management in dentistry	Infection control in dental practice and occupational hazards ● Communicable & Non-communicable Disease in related to dentistry ● Sterilization ● Safety measure in dental practice	L=2 T=5

Learning Objectives and Course Contents in Dental Jurisprudence

Learning Objectives	Contents	Teaching Hours
Student will be able to Define ethics Principles of ethics Geneva declaration Demonstrate the knowledge and understanding of ● Civil Law ● Criminal Law ● Legal process ● Define and classify negligence ● Define and classify assault ● Define and classify consent ● Define and classify defammation ● Res ipsa loquitor	Law & Ethics in dentistry	L=2 T=4

Learning Objectives	Contents	Teaching Hours
- Describe ordinance & regulations relating to medical & dental practice. - State legal rights & protection	- The acts and ordinance relating to Medical & Dental practice and Drug regulation - Legal rights & protection	L=1 T=2
- Define record keeping. - Rules of record keeping - Importance and Use of record keeping	Dental record keeping	L=1 T=2
Mention the organizer & functions National Health Services, BMDC& National Dental Organization. WHO, FDI, BDS , etc.	Organizations. National & International Health Organization	L=2 T=4
- Define forensic odontology - Explain the scope of Forensic Odontology - Importance of Forensic Odontology. - Describe Process of person identification - Age determination - Sex determination - Bite Mark registration - Function of forensic odontologist	Forensic odontology	L=2 T=4

Learning Objectives and Course Contents in Behavioral Science

Learning Objectives	Contents	Teaching Hours
Student will be able to ● Define behavioral science & mention its components; explain its scope & uses in dental health care. ● State types of behavior ● Define cognition ● Explain personality ● Define motivation ● Manage stress ,fear and anxiety in dentistry	Concept of Behavioral Science	L=2 T=4
● Define Society, Culture ● Enumerate components of culture ● Identify various social & culture factors which influences health	Social, Cultural and psychological factors in health and illness	L=1 T=2
● Define interpersonal relationship ● Mention factors influencing interpersonal relationship & doctor-patient relationship & the technique to improve such relationship. ● Types of doctor patient relation	Interpersonal relationship Doctor – patient relationship Doctor- nurse relationship	L=1 T=2

Learning Objectives	Contents	Teaching Hours
Student will be able to ● Define Family ● Mention types of Family ● Family cycle ● Basic need of family ● Describe role of family in health & illness	Concept of Family	L=1 T=2
● Describe Illness behavior & management	Illness behavior and its management	L=1 T=2
● Define communication ● Mention components of communication ● Classify types of communication ● Classify types of media ● Mention barriers in communication ● Describe behavior changing process ● Describe steps of Behaviour change communication	Behavioral Change Communication (BCC)	L=1 T=3